

Tide retrospective: the q -uniform window majorant the single interface of the dominated-convergence step

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The architecture consult called this theorem the numerator stage’s “key hidden dependency”: everything the dominated-convergence step needs about the integrand, in one statement. On the mesoscopic window, eventually in q ,

$$\left| e^{-(L(qz)-L(0))/q^2} - e^{-T_2(z)} \sum_{j=0}^N P_j(z) q^j \right| \leq C (1 + \|z\|)^{(N+2)(N+1)} q^N e^{-(\lambda/4)\|z\|^2},$$

with C explicit: the domain’s Taylor-window constant, plus $M^{N+1}/(N+1)!$ from the exponential remainder, plus the tail coefficient sum through degree N^2 . The proof factors the true Boltzmann factor through the exponent split — whose critical-point hypothesis is now the bridge theorem of the previous tide, not an assumption — and telescopes the difference into the three scalar pieces prepared in 5b: the perturbation strip carrying $|q^N \rho_q(z)|$, the unrestricted exponential Taylor remainder at $-A(q, z)$ with the multinomial extraction $|A|^{N+1} \leq q^{N+1} (\sum_s |V_s(z)|)^{N+1}$, and the graded polynomial tail carrying q^{N+1} . Every exponential factor absorbs into the weakened Gaussian of 5c-pre.

1 Setting

Stage 5c-i of the forward-expansion programme. The four preceding tides were built so that this one would be assembly: 5a supplied the coefficient growth, 5b the three scalar estimates, 5c-pre the bridge and the absorption. The assembly is still the longest single proof of the programme (one theorem, ~ 260 lines), but it compiled on the second build attempt with six local fixes and no restructuring — the decomposition did its job.

2 The thing formalised

`sum_range_shift_eq_sum_Icc` aligns the exponent split’s $\sum_{s \in \text{range } N} q^{s+1} V_{s+1}$ with the graded expansion’s $\sum_{s \in \text{Icc } 1 \ N} V_s q^s$. The majorant `normalized_window_remainder_bound` then: obtains the absorption, ball-landing, and $q \in (0, 1)$ facts eventually; factors via `exponent_split` discharged by `taylorHomogeneousTerm_one_eq_zero`; sets S (the homogeneous correction), R (the scaled perturbation), and the graded polynomial G ; and bounds the telescope $|e^{-(S+R)} - \Sigma| \leq |e^{-(S+R)} - e^{-S}| + |e^{-S} - G(q)| + |G(q) - \Sigma|$ piece by piece. The pieces reuse, in order: `abs_exp_neg_add_sub_exp_neg_le` with `abs_scaledRem_le`; `abs_exp_sub_sum_le` with `abs_exponent_sum_le` and the growth bound $M(1 + \|z\|)^{N+2}$ on

$\sum_s |V_s|$; and `gradedExpPoly_tail_bound` with the 5a coefficient growth. Absorption closes each piece at rate $\lambda/4$.

3 Where progress stalled

Six small fixes on one round, all catalogued classes or new catalogue entries: `add_le_add_right` adds on the left in this Mathlib (use `add_le_add h le_rfl`); `positivity` cannot see nonnegativity of opaque structure fields (derive a local fact for `remConst` once, thread it); `Metric.mem_ball_zero_iff` does not exist on this pin (rewrite through `Metric.mem_ball` and `dist_zero_right`); a `gcongr` discharged its own side goal; and — the one genuinely structural point — the `set ... with` folding of R must happen *after* the absorption instance is obtained, so the fold reaches into that hypothesis and `linarith`-style steps see one atom.

4 Mathlib lookup versus new mathematics

Lookup: `Finset.sum_Icc_succ_top`, `Metric.mem_ball`, `abs_sub_le`, and the mono/absolute-value toolkit. Everything else was this programme’s own five preceding stages. New mathematics: none — the theorem is the disciplined sum of its parts, which was the design.

5 What the next tide inherits

Stage 5c-ii, the final integration: pointwise convergence of the normalized window remainder (`exp_graded_expansion` at $a_s = V_s(z)$, ρ = the scaled remainder, activated by `eventually_mem_mesoscopicSet`); the filter-indexed dominated convergence with the window indicator folded into the integrand and this tide’s majorant as the dominator (integrable by `integrable_one_add_norm_pow_mul_gaussian`); the two outer-tail removals from stage 2; coefficient integrability; and the packaging of `numerator_hasExpansion`.